Padmanava Karmakar

AI Flutter Engineer | 4.5+ Years | Flutter, Dart & AI-Assisted Development (Claude Code, Cursor, Antigravity)

Ahmedabad, Gujarat, India · Open to relocation · Immediate joiner

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Professional Summary

AI Flutter Engineer with 4.5+ years building cross-platform iOS and Android apps in Flutter and Dart across healthcare (HIPAA), e-commerce, logistics and real-time location domains. Strong in Clean Architecture, MVVM, Bloc/Riverpod/GetX state management, Firebase, Supabase, REST API integration and App Store/Play Store releases, and has integrated Google Gemini AI features into Flutter apps.

Advanced practitioner of AI-assisted development with Claude Code and Cursor. Automated an end-to-end ticket workflow with Claude Code (pick up, implement, test, update, close) that resolved 50–60 tickets per day for several months on a deadline-critical client project. Led the mobile module of a three-portal telemedicine platform, and runs internal knowledge-sharing sessions.

Technical Skills

- **Languages:** Dart, Python
- **Mobile:** Flutter 3, Material Design 3, Responsive/Adaptive UI, Custom Animations, Platform Channels, Flutter Web, In-App Purchases, Localization (i18n), Deep Linking, Push Notifications, Offline-First Sync, Video Calling (Jitsi Meet SDK), Isolates, Build Flavors, App Performance Optimization
- **State Management & Architecture:** Bloc, Cubit, Riverpod, Provider, GetX, Clean Architecture, MVVM, SOLID, Dependency Injection (GetIt), GoRouter, Code Generation (build_runner)
- **AI-Assisted Development:** Claude Code, Cursor, AI-driven ticket automation (implement, test, update, close)
- **AI Integration:** Google Gemini API, ChromaDB, Typesense semantic search, ML models (XGBoost, LightGBM, SHAP)

- **Backend & APIs:** RESTful APIs, dio, Retrofit, WebSockets, Node.js, Express, FastAPI, JWT, Firebase, Supabase
- **Databases & Cloud:** Firebase (Firestore, Auth, Storage, Cloud Functions, FCM, Remote Config, Analytics), Supabase, PostgreSQL, Redis, SQLite/SQFlite, Isar
- **Maps & Location:** Google Maps, Google Places API, Mapbox, OLA Maps, real-time location tracking
- **Payments & Security:** Stripe, End-to-End Encryption (RSA/AES), Flutter Secure Storage, MFA, HIPAA compliance
- **Testing & DevOps:** Unit, Widget & Integration Testing, CI/CD (GitHub Actions, Codemagic, Fastlane), Firebase Crashlytics, Git/GitHub, Trello, App Store & Play Store Deployment

Professional Experience

Software Engineer — Bacancy Technology, Ahmedabad

Sep 2023 – Present

- Led the mobile module of a HIPAA-compliant telemedicine platform as sole Flutter developer, building 50+ screens across Patient, Doctor and Admin portals with Riverpod, GoRouter, dio and flutter_secure_storage, and integrated the Jitsi Meet SDK for doctor–patient video consultations.
- Adapted the platform's existing Next.js web APIs for mobile as one of four developers on the mobile API layer, making them compatible with the Flutter app's auth, patient, doctor and admin flows.
- Delivered the appointment-booking flow, from doctor and slot selection to Stripe checkout, plus role-based authentication with email-OTP MFA and idle-session auto-logout; configured dev/prod build flavors for iOS and Android.
- Automated the ticket lifecycle with Claude Code (pick up, implement, test, update and close tickets), resolving 50–60 tickets per day for several months on a deadline-critical client project.
- Built core features of The Word, a social-impact app (Flutter, Firebase, Riverpod), in a 3-developer team: end-to-end encrypted chat (RSA/AES), real-time messaging with Firestore and Cloud Functions, Stripe donations, Typesense search, Google Places discovery and Google Gemini AI features.

- Revamped Goldbank Driver, a London logistics app, with Riverpod and Mapbox real-time journey tracking synced to the admin panel; shipped multiple production releases of Goldbank Live, live on the App Store and Play Store, and maintained Sell Gold (GetX) through UI updates, API integrations and bug fixes.
- Shipped App Store and Play Store releases through CI/CD pipelines (Codemagic, GitHub Actions, Fastlane), backed by unit, widget and integration tests and Firebase Crashlytics monitoring.
- Led the team's Stack Overflow question-solving group, ran internal technical sessions, mentored juniors, and handled code reviews, estimations and client communication.

Associate Software Engineer — Radixweb Pvt Ltd, Ahmedabad

Sep 2022 – Aug 2023 · Trainee Software Engineer, Feb 2022 – Aug 2022

- Built DigInn, a healthy-food ordering app (pickup and delivery) live on the App Store and Play Store, and shipped multiple production releases using GetX and Clean Architecture in a team of 2 developers, a QA engineer and a tech lead.
- Developed meal customization with full nutrition details and a three-tier loyalty program (Bronze, Silver, Gold) with sign-up and birthday rewards.
- Integrated REST APIs with dio and Retrofit, the Thanx SDK for deep links, Firebase Remote Config, Analytics and FCM, and Isar for local storage.
- Created reusable custom widgets, light/dark theming and shared text styles, and wrote unit tests.
- Built a news app during training using Cubit, dio and Clean Architecture, with category and country filters and keyword search with suggestions.

Personal Projects

Built end to end using AI-assisted development (Claude Code, Cursor).

Cafeasy — Location-Aware RAG Recommendation Engine

Python, FastAPI, ChromaDB, LLMs, Vector Embeddings

- Built a location-aware RAG recommendation engine running hybrid vector search over scraped cafe data, served through a FastAPI backend.
- Grounded LLM responses only in retrieved records, using dense vector embeddings, to prevent hallucinated places; every recommendation carries a verified Google Maps link and appears on a real-time map.

AstraQuant — AI Stock Intelligence App (NSE/BSE)

Flutter, Riverpod, GoRouter, fl_chart, FastAPI, Python, XGBoost, LightGBM, SHAP, WebSockets, Redis, PostgreSQL

- Built ML-powered stock predictions from a meta-learner ensemble of technical, fundamental, sentiment and market-regime models, with SHAP explanations for every prediction.
- Streamed real-time market data over WebSockets with Redis caching and PostgreSQL persistence; added news sentiment analysis, model drift alerts and portfolio tracking across stocks, ETFs, mutual funds and commodities.

Alankar — AI Music Education App

Flutter, Provider, flutter_soloud (C++), FFT, Supabase, SQLite, Google Gemini

- Built real-time FFT-based pitch detection with sub-cent accuracy and a Gemini-powered vocal coach that explains each suggested correction.
- Created a low-latency harmonium with bellows physics, plus tanpura and tabla, 40+ raags and 100+ sargam patterns, offline-first SQLite storage with Supabase sync, and streak heatmaps.

Santa.com — AI-Powered E-Commerce Platform

Flutter, Riverpod, GoRouter, Supabase, Firebase, Node.js, Express, JWT, PostgreSQL, React, Vite, Tailwind

- Built a full-stack ecosystem: Flutter shopping app, Node.js/Express API with JWT auth, and a React admin panel.
- Shipped visual search, an AI stylist, AR try-on, social Squad Cart and price-haggle mode, plus admin analytics, vendor KYC, audit logs and 2FA.

Rider Buddy — Real-Time Group Cycling App

Flutter, Firebase, Google Maps

- Built live group location sharing with ~10 m accuracy, QR-based group management, member status tracking and meeting-point suggestions for late joiners.
- Optimized Firebase listeners to support up to 1,000 concurrent users, with marker clustering, caching and offline sync for fast map rendering and low battery use.

Education

- **Master of Computer Applications (MCA)** — Parul University, Vadodara · 2022 · CGPA 8.3/10
- **B.Sc. Physics (Honours)** — Krishnath College, West Bengal · 2019

Languages

English, Hindi, Bengali, Gujarati